Wang 2019 Glioblastoma — analysis summary

RUN 1 OF 8

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

7/10 samples (≥237 cells) intermix on one continuous manifold; no per-patient islands (16 detached cells removed); Harmony drops same-sample neighbour fraction 0.99→0.63, iLISI 1.01→2.00.

Gate 2 — module geography → GO (non-branching)

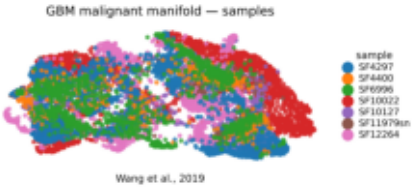
Single developmental axis, not a bifurcation: Gast & pSom co-vary weakly (r=+0.15), Crest is the inverse terminal gradient (vs pSom r=−0.87; Crest-high decile at pseudotime 0.86). No competing arms.

Gate 3 — hub cross-repression → NA

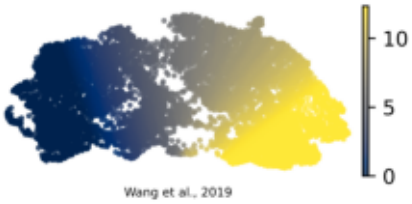
Branching-only test. Non-branching manifold has no Gast/pSom hub bifurcation, so the mutual-exclusivity correlation is not defined here.

STEP FIGURES

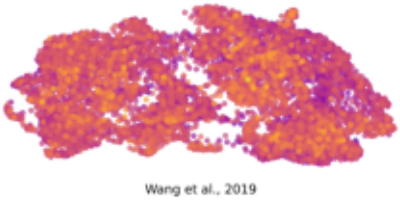
Manifold — samples



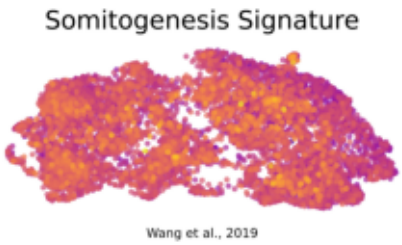
Pseudotime



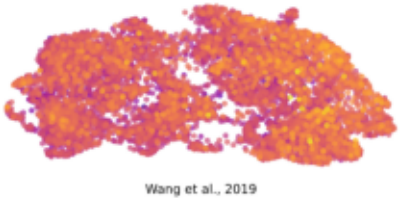
Gastrulation



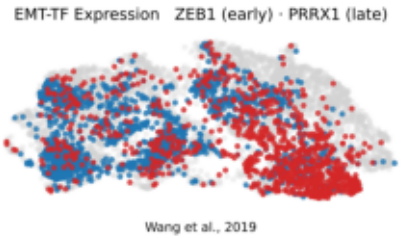
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Glioblastoma (Wang et al., 2019), 10x — 10 patients / 16,317 cells (master scope); 9,916 malignant cells, 9,634 after thresholding thin samples and removing detached islands.

Verdict.
Pattern reproduces (non-branching). A single embryonic axis runs from a gastrulation/somitogenesis-high origin to a neural-crest-high terminus; pseudotime is developmental, not proliferative (cycle-score rho +0.02). EMT-TF axis ZEB1 (root) → PRRX1 (terminus).

Parameter choices

Min cells/sample	≥237 (largest multiplicative gap; keeps 7/10)	**PC count**	30 (chord-elbow 8 over-cuts — Chen2020 lesson)
Integration	Harmony on sample, θ=10 (mixing plateau)	**Island removal**	UMAP radius-graph, −16 cells / 1 island
Trajectory	scFates curve N=20 λ=0.3 μ=100 (fold-free, ratio 2.87)	**Root**	curve tip 2 = pSom/Gast-high, Crest-low
Subtypes	3CA meta-programs (Neftel: Astro / MES-Hypoxia / OPC) + Wang 2019	**Geometry**	Linear-pSom/Crest (single axis; pSom strongest globally)